

CS585 Final exam

2025-05-07; duration: 1 hour

Name (Last, First): _____

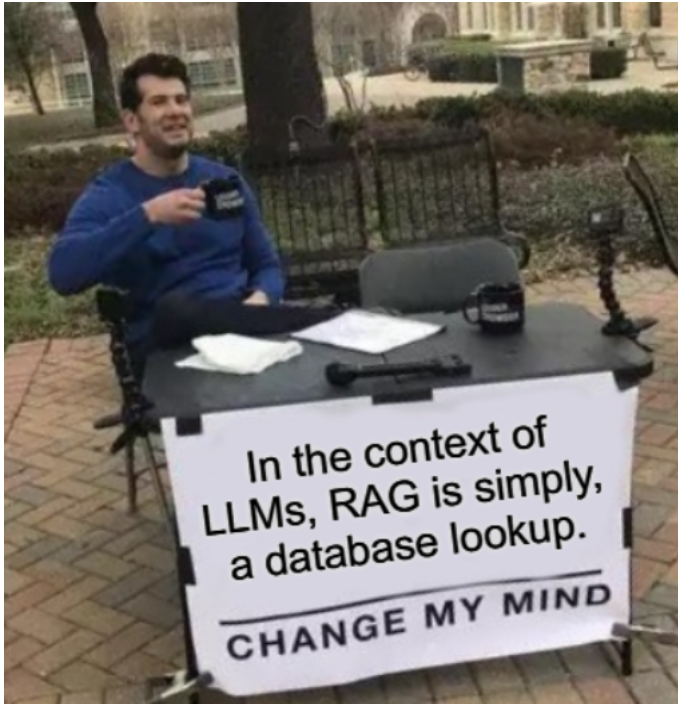
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Please read the five items below, carefully - it's how we roll :)

1. There are 10 questions, each worth 3 points. Please just answer the question, don't write 'more' (ie. provide irrelevant info) - no points for writing 'more'.
2. Please **STOP** writing when the time is up, then upload your exam. You will **NOT** get additional time - if you are late submitting, you will get a **5 point** penalty.
3. You can use your cheatsheet, **nothing else**. If you are observed cheating, you'll get a 0 for the exam, **AND**, be reported to OAI [fyi: we report first, **THEN** notify you]. Please ***DO NOT*** cheat.
4. Every question has 'data' or 'database' in it :)
5. Have fun!

Q1 (1+1+1 points).



Explain the above. Specifically, we build a kind of 'database':

- what is it (the database) called?
- what does it store?
- how do we retrieve?

Q2 (1+2 points).

v1 of LLMs is simply, prompt-based response.

v2 involves RAG.

v3 is 'agentic'.

a. What does 'agentic AI' mean?

b. In addition to the LLM being a core aspect of an agent (because we use it for natural language-based interaction with the agent), what are two other components that are used by (needed for) an agent? Hint - one is 'workspace', another is 'backend' [can include database operations] :)

Q3 (2+1 points). In terms of connectivity (ie. a frontend to a DB via a server), REST and GraphQL are two important, widely used protocols (eg REST API calls are routed through the server to fetch data to the client, eg. weather, stock price). Agentic AI adds another “layer” on top of these (a new standard being widely and quickly being adopted).

a. EXPLAIN (what it is called, what makes it useful (ie. what problem does it solve).

b. the layer referenced above, resembles a construct in MapReduce processing (hint: related to frontend ‘graphs’, backend ‘processors’) - what is it?

Q4. Here is a recent exchange between a judge, and Meta:

"But if I'm going to steal things from the marketplace of ideas in order to develop my own ideas, that's copyright infringement, right?" Chhabria responded.

[the judge]

Wired noted that he asked Meta's lawyers, "What about the next Taylor Swift?" If AI made it easy to knock off a young singer's sound, how could she ever compete if AI produced "a billion pop songs" in her style?

In a statement, Meta's spokesperson reiterated the company's defense that AI training is fair use.

"Meta has developed transformational open source AI models that are powering incredible innovation, productivity, and creativity for individuals and companies," Meta's spokesperson said. "Fair use of copyrighted materials is vital to this." We disagree with Plaintiffs' assertions, and the full record tells a different story. We will continue to vigorously defend ourselves and to protect the development of GenAI for the benefit of all."

'Fair use' claim is quite dubious ("sus"!) in this context. Why? In other words, Meta's claim is ***unjustified***, as are similar claims by OpenAI, Midjourney, and Perplexity AI (which claimed that 'news' belongs in the public domain) [they all claim the right to use copyrighted data for model training]. Discuss the reasons why (it's not justifiable).

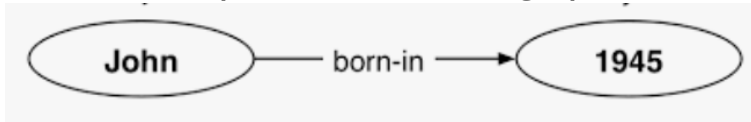
Q5. As we saw, neo4j is a leading graph database company [whose graphs can be queried via Cypher, or TinkerPop].

Such graphs have a 'newfound' purpose - in what else - for use with LLMs :)

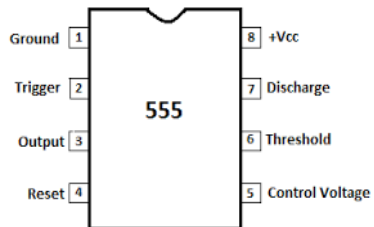
How so? Discuss [using a simple 'systems' diagram if you like]. HINT - this is related to Q1 and Q2 above.

Q6 (1+1+1 points). Express the following data as **valid** JSON.

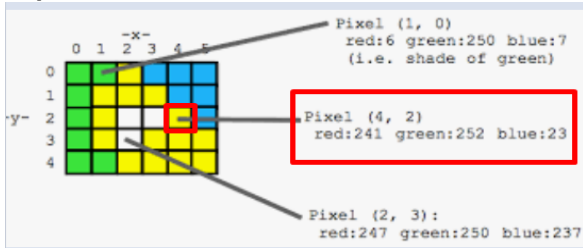
A semantic triple (remember our graph DB lecture?):



A timer IC:



A pixel [JUST the one outlined in red]:



Q7 (2+1 points).

Database columns need to be INDEXED, for fast retrieval of data in those columns (eg. stock price data, to use a numerical example; other types such as strings and dates can be indexed too, as you know).

a. How does indexing reduce lookup time from $O(N)$ to $O(\log N)$? Explain, using a simple diagram. N stands for column size, ie. row count.

b. In certain cases, it might be possible to reduce the search to $O(1)$: ultra-fast lookup not dependent on column size (ie. N)! How?

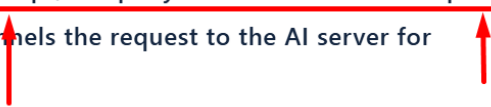
Q8. 'BI' was the beginning of 'data driven' insights - mine a typically large volume of data, to understand past outcomes and to predict future values.

'AI', notably *supervised* machine-learning, also depends on data, to display intelligent behavior. How? Explain, using a simple diagram.

Q9. How is spatial data indexing, related to a type of indexing used in conjunction with LLMs (HINT: Q1!!!). Explain BOTH (spatial, the LLM-related one) using simple diagrams, keeping the answers short and to the point.

Q10.

How semantic caching works:

1. A user sends a chat to an LLM-based AI agent.
 2. A layer intercepts the request and performs a semantic analysis to determine if the request matches any previously made prompts.
 3. If the system finds a suitable prompt, it rapidly returns the cached response.
 4. If no similar prompt exists, it funnels the request to the AI server for processing.
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In the above, 'semantic caching' (aka CAG - Cache Augmented Generation, rhymes with RAG, lol) is used to rapidly return an item IF it's been cached.

In performance tuning of relational DBs, a very similar caching mechanism is used (we studied this) to speed up data processing. Explain.